

Unified Cyber-Physical Sensing & Agentic AI Flash Flood Forecasting

Multi-Scale Flood Forecasting: Combining ESP32 Ground Node, Open-Meteo Satellite NWP & TreeSHAP Explainable ML

Document: **Master
Architecture Report v2.4**

Classification: **Smart India Hackathon (SIH
2026) / Research Whitepaper**

Status: **System
Operational & Calibrated**

1 Purpose & System Vision

PravahAI integrates two complementary data streams to provide reliable, multi-scale flood forecasting and life-saving early warning:



Macro / Regional Layer

Open-Meteo & IMD NWP ensemble precipitation, 16-day synoptic projections, and SRTM 30m Digital Elevation Models (DEM).



Micro / Local Layer

ESP32 cyber-physical sensor node capturing real-time ground wetness, river gauge depth surge, and volumetric soil saturation.



Multi-Source Fusion

Dual XGBoost inference + TreeSHAP explainability engine + OSM Dijkstra safe evacuation route & shelter allocator.

2 Technical Honesty & Jury Defense Strategy

⚠️ Mandatory Evaluation & Hackathon Defense Rule:

Low-cost physical analog probes (e.g. 2 cm raindrop sensors) measure surface conductivity and electrical resistance, not calibrated physical millimetres in free atmosphere. In PravahAI, we explicitly label these measurements as **"Ground IoT Observation (Simulated Rainfall Equivalent Index)"**.

Anticipated Jury Question	Defensible Technical Answer
"Aapka 2 cm ka rain sensor 200 mm real rainfall kaise measure kar raha hai?"	Physical sensor ground-level wetness aur electrical resistance measure karta hai. Humara Calibration Engine use normalized index ($0-100\%$) me convert karta hai aur trained catchment hydrology curve ke hisaab se catchment-equivalent precipitation ($R_{3d} \leq 320 \text{ mm}$) estimate karta hai. UI par ise clearly 'Simulated Equivalent Index' label kiya gaya hai.
"Agar Satellite Weather aur Local Sensor me difference ho toh model kya karega?"	Humne ' Compare Results (Discrepancy Engine) ' diya hai — agar satellite dry dikhata hai lekin ground sensor sudden channel surge detect karta hai, toh Agentic AI instantly Localized Flash Flood Warning trigger karta hai without waiting for satellite orbital pass.

3 Hardware Configuration & ESP32 Pinout

Component / Sensor	Sensor Output	ESP32 Pin	Operating Voltage	Role in Demonstration
ESP32 NodeMCU (WROOM-32)	Wi-Fi 802.11 b/g/n	Microcontroller	3.3V / 5V	Reads analog/digital sensors and transmits JSON payload via HTTP POST.
Raindrop Sensor Module	Analog Out (AO)	GPIO 34 (ADC1_CH6)	3.3V / 5V	Detects surface mist/water spray to simulate precipitation accumulation.
Water Level Sensor Probe	Analog Signal (S)	GPIO 35 (ADC1_CH7)	3.3V / 5V	Measures water depth when submerged in a water container.
Soil Moisture Sensor	Analog Out (AO)	GPIO 32 (ADC1_CH4)	3.3V / 5V	Measures soil moisture saturation in demo pot.
DHT11 Sensor	Digital Serial	GPIO 4 (Digital I/O)	3.3V / 5V	Captures ambient temperature and relative humidity.

4 Mathematical Physics & Calibration Curves

The Python Master Backend converts raw 12-bit ADC values (\$0-4095\$) into domain hydrological features:

A. Rainfall Scaling (Dry ≈ 4095 , Saturated Wet ≈ 400 ADC):

```
Wetness_Ratio = max(0.0, min(1.0, (4095 - Raw_Rain) / 3600))
Rain_Index_% = Wetness_Ratio * 100%
Scaled_R3d = Wetness_Ratio * 320.0 mm
Scaled_R1h = Wetness_Ratio * 35.0 mm/h
```

B. River Catchment Depth & Submersion Relative to Danger Benchmark:

```
Submerged_Ratio = max(0.0, min(1.0, (Raw_Water - 300) / 3200))
River_Gauge_Level (H) = 20.00m + (Submerged_Ratio - 0.40) * 1.60m
Difference_to_Danger = H - 20.00m
River_Discharge (Q) = 220 + (Submerged_Ratio * 1400) m³/s
```

C. Soil Moisture Saturation Index:

```
Soil_Wetness_Ratio = max(0.0, min(1.0, (4095 - Raw_Soil) / 3200))
Soil_Saturation_% = min(98.0%, 35.0% + (Soil_Wetness_Ratio * 62.0%))
```

5 Triple-Action Workflow & Dual Inference

Prediction Action	Data Source Used	Primary Use-Case
[🌐 Predict Using Weather]	Open-Meteo + IMD NWP Forecast	Macro-regional forecasting up to 16 days in advance across state basins.
[⚡ Predict Using IoT]	ESP32 Ground Physical Node (Live)	Immediate catchment runoff and rapid river stage surge evaluation.
[⚖️ Compare Results]	Dual Satellite vs IoT Inference	Detects ground-flash discrepancies (ΔRisk) and generates explainable early advisories.

6 End-to-End Test Scenarios & Validation

Scenario	Physical / Simulator State	Scaled Hydro Metric	ML Output & Alert Action
Scenario 1: Baseline Dry	Dry sensor, dry soil, water probe in air	Rain: 0 mm, Soil: 35%, River: -0.64m below danger	Low Risk (15-25%) — Normal Monitoring
Scenario 2: Rain Spray Surge	Water spray mist on raindrop sensor	Rain: 210 mm, Soil: 68%, River: Safe depth	Moderate/High Risk (65-75%) — Advisory
Scenario 3: Extreme Flash Flood	Soaked sensor + submerged probe + wet soil	Rain: 320 mm, Soil: 97%, River: +0.96m above danger	Very High Risk (99%) — Evacuation & CAP XML